

BlockerAnalyzer — Per-Shape Hypothesis-Cluster Triage

Source: bench/dataset.jsonl (622 rows = branch × shape axes over 554 distinct branches; 330 validated / 292 inconclusive rows). #br = validated branches the cluster owns. * = unbuilt tool. [df] = decidable:false.

Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
afflast_rarity_LW	afflast_nfuzz_damping_starves_deep_path [df]	AFLFast FAST n_fuzz rarity damping de-prioritizes a frequently-hit (saturated) seed and starves the deep path of sustained energy; minimizer's calibration-only schedule (PowerSchedule None) keeps allocating mutations and reaches it	0	energy_share*	fast_energy_share_on_common_edge_seed, minimizer_energy_share_on_common_edge_seed, fast_depth_reach, minimizer_depth_reach	fast_energy_share_on_common_edge_seed : fraction of havoc energy/execs the afflast(fast) schedule spends on the rare-edge seed minimizer_energy_share_on_common_edge_seed : fraction of havoc energy/execs the minimizer schedule spends on the rare-edge seed fast_depth_reach : max coverage-edge depth past the rare edge reached by afflast(fast) descendants minimizer_depth_reach : max coverage-edge depth past the rare edge reached by minimizer descendants	DECIDABLE:FALSE – no rule can fire. fast_energy_share_on_common_edge_seed and "depth_reach are NOT obtainable: this is a runtime scheduling dynamic (campaign-wide n_fuzz history + per-seed energy...
afflast_rarity_LW	(inconclusive)	branches with no validated label	2	—	—	—	—
afflast_rarity_WL	rare_edge_energy_overdrive [df]	fast over-invests havoc energy in the rare-edge seed, so its corpus shows more exploration past the rare edge than minimizer's	0	energy_share*	rare_seed_descendant_share, rare_seed_depth_reach	rare_seed_descendant_share : fraction of the afflast(fast) corpus that descends from the rare-edge seed rare_seed_depth_reach : max coverage-edge depth past the rare edge reached by afflast(fast) descendants	rare_seed_descendant_share ≥ 1.5 AND rare_seed_depth_reach > 0
afflast_rarity_WL	(inconclusive)	branches with no validated label	2	—	—	—	—
calibrated_energy_WL	calibrated_structural_assembly	Calibrated energy concentrates havoc on compact/fast/stable seeds, so its corpus assembles a LARGER / more-structured (more-token) input than naive's flat budget ever funds	0	joint_necessity	size_lift, tag_lift	size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows) tag_lift : winner_tags — loser_tags (structural-token advantage)	size_lift ≥ 1.3 AND tag_lift ≥ 1.3
calibrated_energy_WL	calibrated_rare_value_targeting	Calibrated energy raises per-seed havoc count on near-valid seeds, raising the probability of landing a specific RARE byte / multi-byte codepoint value at the gating position; the win is value-precision, NOT a larger structure	0	value_distance_reached	value_dist_lift, calibrated_hits_value	value_dist_lift : winner/loser ratio of value-distance closed calibrated_hits_value : whether the calibrated-energy arm hits the rare target value at all	calibrated_hits_value == true AND value_dist_lift > 1.0
calibrated_energy_WL	(inconclusive)	branches with no validated label	13	—	—	—	—
ctx_coverage_LW	ctx_corpus_inflation	ctx inflates the corpus / map so naive_ctx hoards far more (context-redundant) entries than naive	2	corpus_size_ratio	corpus_count_ratio	corpus_count_ratio : saved-corpus size (or l2S-arm/baseline size ratio) — corpus bloat	corpus_count_ratio ≥ 1.5
ctx_coverage_LW	ctx_depth_inflation_at_gate	a high-call-count / multi-context path at the gate is what ctx fragments (flooding) vs a shallow exact-literal gate (dilution)	2	depth_reach	call_count_lift	call_count_lift : naive_ctx/naive ratio of calls into the gate function (loop flooding vs dilution)	call_count_lift ≥ 3.0 (flooding sub-mechanism) vs call_count_lift < 3.0 (dilution sub-mechanism)
ctx_coverage_LW	(inconclusive)	branches with no validated label	2	—	—	—	—
ctx_coverage_WL	iteration_depth_threshold	naive_ctx accumulates higher loop/recursion iteration depth at the blocker than naive, clearing a count/nesting threshold	14	depth_reach	iter_depth_lift, naive_ctx_max_iter, naive_max_iter	iter_depth_lift : naive_ctx / naive ratio of max loop/recursion iteration count at the gate naive_ctx_max_iter : max loop/recursion iteration count at the gate in the naive_ctx arm naive_max_iter : max loop/recursion iteration count at the gate in the naive arm	iter_depth_lift ≥ 1.5 AND naive_ctx_max_iter > naive_max_iter
ctx_coverage_WL	context_reach_advantage	naive_ctx's resolving corpus reaches the blocker through call-contexts / path-depths that naive's blocking corpus never reaches	0	depth_reach	ctx_reach_lift, naive_ctx_distinct_contexts, naive_distinct_contexts	ctx_reach_lift : winner/loser ratio of seeds (or contexts) reaching the blocker naive_ctx_distinct_contexts : # distinct call-contexts the naive_ctx corpus reaches the blocker from naive_distinct_contexts : # distinct call-contexts the naive corpus reaches the blocker from	ctx_reach_lift ≥ 1.5 AND naive_ctx_distinct_contexts > naive_distinct_contexts
ctx_coverage_WL	(inconclusive)	branches with no validated label	39	—	—	—	—
grimoire_structural_LW	gap_erases_exact_literal	Grimoire's GeneralizationStage replaces the concrete literal byte (l2S/havoc planted) with a <GAP> token; structural recombination never restores it, so a byte-equality gate that literal-preserving naive can satisfy stays blocked under grimoire.	0	token_count	naive_literal_count, grimoire_literal_count, literal_presence_ratio	naive_literal_count : # of the exact gate literal occurrences in naive seeds grimoire_literal_count : # of the exact gate literal occurrences in grimoire seeds literal_presence_ratio : fraction of winner seeds still carrying the exact literal after the mutator	naive_literal_count ≥ 1 AND literal_presence_ratio < 0.34
grimoire_structural_LW	fragmentation_starves_depth	Grimoire's re-run tax + structural-recombination produce short, fragmented inputs that never advance the parser far enough to pass a quantitative buffer/depth threshold; naive's longer coherent content satisfies the buffer-ahead condition.	0	joint_necessity	naive_median_size, grimoire_median_size, size_lift	naive_median_size : naive median seed length grimoire_median_size : grimoire median seed length size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows)	size_lift ≥ 3.0 AND grimoire_median_size < naive_median_size
grimoire_structural_LW	(inconclusive)	branches with no validated label	3	—	—	—	—
grimoire_structural_WL	structural_token_assembly	grimoire's grammar mutators place structural tokens/constructs into the corpus that naive's byte havoc cannot, building larger/deeper inputs	9	joint_necessity	naive_tags, grimoire_tags, tag_lift, size_lift	naive_tags : mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the naive arm grimoire_tags : mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the grimoire arm tag_lift : winner_tags — loser_tags (structural-token advantage) size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows)	tag_lift ≥ 1.0 AND naive_tags < grimoire_tags AND grimoire_tags ≥ 2
grimoire_structural_WL	structural_depth_only	fallback sub-hypothesis: grimoire reaches the gate via larger/deeper structure without a target-vocabulary token signal (size lift but flat token count)	2	joint_necessity	size_lift	size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows)	size_lift ≥ 1.3
grimoire_structural_WL	(inconclusive)	branches with no validated label	31	—	—	—	—

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Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LLWL	joint_assembly_depth	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value_profile) places ~0 structural tokens	6	joint_necessity	vp_tags, size_lift	vp_tags: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the value_profile arm size_lift: winner/loser median seed-size ratio (assembly extent the gradient grows)	vp_tags < 0.3 AND size_lift ≥ 1.3
i2s_vp_LLWL	joint_value_precision	vpc and cmplog reach a SIMILARLY-sized structure (size_lift ~ 1, sometimes vpc smaller), but vpc places a specific gradient-climbed VALUE that the cmplog-blocking seeds approach yet never reach; I2S is still necessary to place the surrounding exact literal(s), and the gradient is necessary for the last-mile value precision the structural proxy is blind to	6	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_ratio	vpc_min_distance: min Hamming/value distance from value_profile, cmplog seeds to the gate literal (0 = gate solved) cmp_min_distance: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) distance_closure_ratio: fraction of the value gap the winner closes (vs the loser/naive residual)	size_lift < 1.3 AND cmp_min_distance > 0 AND distance_closure_ratio ≥ 0.5
i2s_vp_LLWL	(inconclusive)	branches with no validated label	11	—	—	—	—
i2s_vp_LLW_	joint_assembly_depth	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value_profile) places ~0 structural tokens	1	joint_necessity	vp_tags, size_lift	vp_tags: mean # structural format-tokens (sfnt/OT tags, PNG chunks, SQL kw...) present per seed in the value_profile arm size_lift: winner/loser median seed-size ratio (assembly extent the gradient grows)	vp_tags < 0.3 AND size_lift ≥ 1.3
i2s_vp_LLW_	joint_value_precision	vpc and cmplog reach a SIMILARLY-sized input (size_lift ~ 1), but vpc lands a specific gradient-climbed VALUE (a single exact multi-byte keyword/token) that the cmplog-blocking seeds approach byte-by-byte yet never reach; I2S is still necessary to splice the keyword operand, and the gradient is necessary for the last-mile prefix precision that the structural size proxy is blind to	9	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_ratio	vpc_min_distance: min Hamming/value distance from value_profile, cmplog seeds to the gate literal (0 = gate solved) cmp_min_distance: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) distance_closure_ratio: fraction of the value gap the winner closes (vs the loser/naive residual)	size_lift < 1.3 AND cmp_min_distance > 0 AND distance_closure_ratio ≥ 0.5
i2s_vp_LLW_	(inconclusive)	branches with no validated label	6	—	—	—	—
i2s_vp_LWLL	vp_gradient_distance_climb	value_profile's CMP_MAP gradient climbs a comparison-operand distance ramp toward a structural/syntactic gate state that cmplog- and naive-blocking seeds never approach	1	value_distance_reached	vp_min_distance, naive_min_distance, cmplog_min_distance, vp_gate_cmp_satisfied_frac, naive_gate_cmp_satisfied_frac	vp_min_distance: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) naive_min_distance: min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) cmplog_min_distance: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) vp_gate_cmp_satisfied_frac: fraction of value_profile seeds satisfying the gate's CMP at the offset naive_gate_cmp_satisfied_frac: fraction of naive seeds satisfying the gate's CMP at the offset	vp_min_distance < cmplog_min_distance AND vp_min_distance < naive_min_distance AND vp_gate_cmp_satisfied_frac ≥ naive_gate_cmp_satisfied_frac + 0.2
i2s_vp_LWLL	(inconclusive)	branches with no validated label	4	—	—	—	—
i2s_vp_LWLW	i2s_decoy_substitution_overfit	I2S substitutes a fixed-offset structural literal (decoy) that pins the loser corpus to the valid-structure path and away from the winner's free-byte target	6	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich: same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich < -0.3
i2s_vp_LWLW	i2s_homogenizes_corpus_starves_state_diversity	I2S operand substitution collapses corpus composition onto a few CMP-matched byte patterns, inflating/homogenizing the I2S-arm corpus so it cannot build the diverse runtime state (linked-list population) the structural predicate needs	0	corpus_size_ratio	corpus_size_ratio, composition_entropy_ratio	corpus_size_ratio: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat composition_entropy_ratio: byte-value Shannon entropy of the I2S-arm corpus \ <i>div</i> baseline (<1 = homogenized)	corpus_size_ratio ≥ 1.3 AND composition_entropy_ratio < 0.85
i2s_vp_LWLW	i2s_no_operand_budget_drain [d1]	Gate is a table-/state-machine-derived or range value with NO fixed-offset input-byte CMP operand; I2S substitution finds nothing to copy and its overhead dilutes the havoc budget that would otherwise land the codepoint by random exploration	0	operand_enrichment	signed_target_enrich	signed_target_enrich: log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	false
i2s_vp_LWLW	i2s_anchors_large_valid_blocks_truncation [d1]	I2S (and ctx) anchor the corpus to large valid-header inputs; the branch needs an input shorter than a 128-byte minimum that only plain havoc byte-deletion produces	0	corpus_size_ratio	median_seed_length_ratio	median_seed_length_ratio: winner/loser median seed length	false
i2s_vp_LWLW	(inconclusive)	branches with no validated label	7	—	—	—	—
i2s_vp_LWL_	vp_gradient_distance_closure	value_profile's CMP_MAP gradient incrementally closes byte-distance to a fixed multi-byte literal gate that cmplog-only and vpc never reach; the gradient (not I2S substitution) is the necessary technique, and adding I2S in vpc diverts energy so it stalls farther from the operand	0	value_distance_reached	vp_min_distance, cmp_min_distance, distance_closure_ratio	vp_min_distance: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) cmp_min_distance: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) vpc_min_distance: min Hamming/value distance from value_profile, cmplog seeds to the gate literal (0 = gate solved) distance_closure_ratio: fraction of the value gap the winner closes (vs the loser/naive residual)	vp_min_distance < cmp_min_distance AND vp_min_distance < vpc_min_distance AND cmp_min_distance > 0 AND distance_closure_ratio ≥ 0.5
i2s_vp_LWL_	(inconclusive)	branches with no validated label	2	—	—	—	—
i2s_vp_LWWL	vp_gradient_value_distance_closed	value_profile's CMP_MAP gradient drives its seeds measurably closer to the gate's target value than cmplog/naive seeds get, while I2S substitution does not close the distance	1	value_distance_reached	vp_min_dist, cmp_min_dist, dist_gap	vp_min_dist: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) cmp_min_dist: min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) dist_gap: winner — loser gap in value/distance closed toward the gate	dist_gap ≥ 0.25 AND vp_min_dist ≤ 0.35 AND cmp_min_dist ≥ 0.50

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Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LWWL	vp_gradient_corpus_reaches_deeper	value_profile's gradient-built corpus reaches a deeper / later parse state than cmplog/naive, corroborating the climb when value-distance is unavailable or the gate is a multi-stage parse depth rather than a single operand	0	depth_reach	vp_depth, cmp_depth, depth_lift	vp_depth : max loop/recursion/path depth reached at the gate by the value_profile arm cmp_depth : max loop/recursion/path depth reached at the gate by the cmplog/I2S arm depth_lift : max loop/recursion/path depth reached at the gate by the arm	depth_lift ≥ 0.20 AND vp_depth ≥ 0.60
i2s_vp_LWWL	(inconclusive)	branches with no validated label	16	—	—	—	—
i2s_vp_LWWW	i2s_anti_structural_byte_corruption	I2S RandReplace overwrites the exact structural/category/literal bytes naive havoc assembles, depleting (not enriching) the target byte in cmplog's corpus relative to naive	7	operand_enrichment	signed_target_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	signed_target_enrich < -0.3
i2s_vp_LWWW	i2s_anti_no_input_preimage_misdirection	Gate has no input-side CMP preimage (category-table / state-machine classification, length threshold, or internally-derived overflow sentinel); I2S substitution finds no fixed-offset value to copy, so it misallocates budget while naive havoc explores freely	0	operand_enrichment	signed_target_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	signed_target_enrich == no_gate_signature
i2s_vp_LWWW	vp_pro_cmpmap_gradient_rescue	VP CMP_MAP Hamming/prefix-distance gradient gives graded partial-progress feedback toward the gate comparand, letting value_profile/vpc retain near-miss inputs that approach the target while cmplog's edge-only binary feedback stalls	1	value_distance_reached	vp_min_value_distance, cmp_min_value_distance, distance_lift	vp_min_value_distance : min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) cmp_min_value_distance : min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) distance_lift : winner—loser gap in value/distance closed toward the gate	distance_lift ≥ 1.3 AND vp_min_value_distance < cmp_min_value_distance
i2s_vp_LWWW	(inconclusive)	branches with no validated label	3	—	—	—	—
i2s_vp_LWW_	vp_gradient_value_distance_climb	value_profile's CMP_MAP Hamming/prefix-distance gradient drives the corpus to smaller residual value-distance at the gating comparison site(s), so vp/vpc reach the comparison target where cmplog (edge-only, no gradient) stalls at a larger distance	0	value_distance_reached	vp_min_distance, cmp_min_distance, distance_closure_ratio	vp_min_distance : min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) cmp_min_distance : min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) distance_closure_ratio : fraction of the value gap the winner closes (vs the loser/naive residual)	distance_closure_ratio ≥ 1.5 AND vp_min_distance < cmp_min_distance
i2s_vp_LWW_	vp_gradient_depth_reach_corroborant	corroborating coverage signal: value_profile/vpc reach a deeper point past the gated comparison earlier than cmplog, target-independently (no corpus needed)	0	depth_reach	vp_reach_frac, cmp_reach_frac, reach_lift	vp_reach_frac : fraction of value_profile seeds that reach the gate region cmp_reach_frac : fraction of cmplog/I2S seeds that reach the gate region reach_lift : winner/loser ratio of seeds (or contexts) reaching the blocker	reach_lift ≥ 0.5
i2s_vp_LWW_	(inconclusive)	branches with no validated label	9	—	—	—	—
i2s_vp_LW_L	vp_gradient_climbs_numeric_threshold	value_profile-resolving seeds drive a derived NUMERIC field past an inequality/range threshold; there is no exact literal to splice, so distance is the MAGNITUDE GAP to crossing the threshold – vp closes it monotonically while cmplog (no single constant to substitute) and naive (no gradient) stall below the bound	4	value_distance_reached	vp_gap, cmp_gap, naive_gap, mag_gap_lift, vp_crossed_frac	vp_gap : value_profile residual gap to the numeric threshold value cmp_gap : cmplog/I2S residual gap to the numeric threshold value naive_gap : naive residual gap to the numeric threshold value mag_gap_lift : winner—loser gap in value/distance closed toward the gate vp_crossed_frac : fraction of value_profile seeds that cross the numeric threshold	vp_crossed_frac ≥ 0.5 AND mag_gap_lift > 0 AND cmp_gap > vp_gap AND naive_gap > vp_gap
i2s_vp_LW_L	vp_gradient_assemblies_multibyte_literal	value_profile-resolving seeds reach a multi-byte operator/keyword/structural LITERAL strictly closer than cmplog-blocking AND naive-blocking seeds; the operand is not a single spliceable constant, so I2S stalls and naive gets no gradient while the CMP_MAP distance bucket ratchets a prefix/Hamming approach to the literal	0	value_distance_reached	vp_dist, cmp_dist, naive_dist, lit_dist_gap, vp_dist_frac	vp_dist : min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) cmp_dist : min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) naive_dist : min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) lit_dist_gap : winner—loser gap in value/distance closed toward the gate vp_dist_frac : value_profile min distance as a fraction of the full gate width	lit_dist_gap ≥ 1.0 AND vp_dist_frac ≤ 0.34 AND cmp_dist > vp_dist AND naive_dist > vp_dist
i2s_vp_LW_L	(inconclusive)	branches with no validated label	3	—	—	—	—
i2s_vp_LW_W	i2s_decoy_substitution_target_depletion	I2S splices a WRONG/competing but abundant CMP operand into the input (a well-known font-table tag, a script-tag/codepoint constant, an in-range fixed-point magnitude, or a competing protocol value), pinning the cmplog corpus onto that decoy and DEPLETING it of the target byte value at the winner gate's offset; naive/value_profile never harvest the decoy so they freely reach the target	23	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	skip != 'no_gate_signature' AND signed_target_enrich < -0.3
i2s_vp_LW_W	i2s_disrupts_havoc_sequence_accumulation	The winner gate needs havoc to BUILD or PRESERVE a long / periodically-repeated glyph-record byte sequence (multi-matra record runs, long multi-glyph Thai state-machine traversal); the gate value is table/state-derived (no harvestable operand), but cmplog's CMP-biased budget produces SHORTER, heterogeneous inputs that never reach the required sequence length, while naive's byte-expansion/copy operators grow and sustain it	0	joint_necessity	size_lift	size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows)	skip == 'no_gate_signature' AND size_lift ≥ 1.3

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Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_LW_W	i2s_overhead_on_derived_gate_no_operand	The gate value is computed from shaping tables, Unicode-property lookups, Rigel state-machine internals, or derived category classification – NOT a literal value compared against input bytes anywhere. I2S has no input-resident CMP operand to harvest or substitute; its per-execution callback overhead and misdirected substitution budget crowd out the broad havoc byte exploration that resolves the branch, with NO directional corpus depletion and NO length asymmetry	0	operand_enrichment	signed_target_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	skip == 'no_gate_signature' AND size_lift < 1.3
i2s_vp_LW_W	(inconclusive)	branches with no validated label	6	—	—	—	—
i2s_vp_L_LW	i2s_case_constant_decoy_substitution	I2S keeps copying enumerated switch case-constants (decoys) into the option-selector byte, pinning control onto a matching case and depleting cmplog/vpc corpora of the non-enumerated values that fall through to default; plain havoc drifts onto default freely	1	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich < -0.3
i2s_vp_L_LW	(inconclusive)	branches with no validated label	0	—	—	—	—
i2s_vp_L_WL	vp_gradient_value_distance_climb	vpc's CMP_MAP gradient ratchets the corpus byte-by-byte toward a CONCRETE literal token that cmplog's edge-only signal cannot bank near-miss progress toward	5	value_distance_reached	w_min_hamming_to_literal, l_min_hamming_to_literal, distance_gap	w_min_hamming_to_literal : (derived metric — see hypothesis prediction/params) l_min_hamming_to_literal : (derived metric — see hypothesis prediction/params) distance_gap : winner – loser gap in value/distance closed toward the gate	distance_gap ≥ 0.25 AND w_min_hamming_to_literal ≤ 0.30
i2s_vp_L_WL	vp_gradient_structural_depth_assembly	vpc's CMP_MAP gradient accumulates partial credit across chained comparisons to drive the corpus to a structurally DEEPER parse state (multi-child content model / namespace binding / internal subset / pointer-identity) whose gate is a derived pointer/state check with NO fixed literal pre-image	0	depth_reach	w_depth, l_depth, depth_lift	w_depth : max loop/recursion/path depth reached at the gate by the winner arm l_depth : max loop/recursion/path depth reached at the gate by the loser arm depth_lift : max loop/recursion/path depth reached at the gate by the arm	depth_lift ≥ 1.5 AND w_depth > l_depth
i2s_vp_L_WL	(inconclusive)	branches with no validated label	9	—	—	—	—
i2s_vp_L_W_	vp_gradient_climbs_multibyte_cmp_beyond_i2s	vpc-resolving seeds reach a multi-byte operand value strictly closer to the target than cmplog-blocking seeds; the operand is not a single spliceable literal, so I2S stalls while the CMP_MAP gradient ratchets distance down	0	value_distance_reached	vpc_dist, cmp_dist, dist_gap, vpc_dist_frac	vpc_dist : min Hamming/value distance from value_profile_cmplog seeds to the gate literal (0 = gate solved) cmp_dist : min Hamming/value distance from cmplog/I2S seeds to the gate literal (0 = gate solved) dist_gap : winner – loser gap in value/distance closed toward the gate vpc_dist_frac : value_profile_cmplog min distance as a fraction of the full gate width	dist_gap ≥ 1.0 AND vpc_dist_frac ≤ 0.34 AND cmp_dist > vpc_dist
i2s_vp_L_W_	(inconclusive)	branches with no validated label	7	—	—	—	—
i2s_vp_L_W	i2s_energy_diversion_decoy_pin	the gate is an ungradable fixed-offset equality; plain havoc places the literal by random mutation while cmplog diverts energy and leaves its corpus depleted of (pinned away from) the target value at the gate offset	2	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich < -0.3
i2s_vp_L_W	i2s_corpus_channeling_starves_target	cmplog solves upstream structural gates and channels the corpus into a deep valid-decode path, starving the shallow target byte of the random mutation plain havoc supplies incidentally	0	corpus_size_ratio	size_ratio_cmp_over_naive, target_reach_ratio_cmp_over_naive	size_ratio_cmp_over_naive : saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat target_reach_ratio_cmp_over_naive : max loop/recursion/path depth reached at the gate by the arm	size_ratio_cmp_over_naive ≥ 1.5 AND target_reach_ratio_cmp_over_naive < 0.5
i2s_vp_L_W	(inconclusive)	branches with no validated label	0	—	—	—	—
i2s_vp_WLWL	i2s_fixed_offset_literal_value_gate	cmplog (I2S) substitutes a logged CMP literal at a fixed gate offset, so its corpus is enriched in the target value at that offset relative to naive; value_profile/naive cannot synthesize the exact literal	73	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 0.8
i2s_vp_WLWL	i2s_multi_literal_scaffold_state_gate [4r]	I2S assembles two or more co-dependent, dispersed CMP literals (URL scheme+port+HTTP tokens; cookie/header chains; deep multi-field ICC structure) whose accumulated downstream program state, not the substitution point itself, controls the blocked branch	0	value_enrichment*	any_anchor_token_enrich	any_anchor_token_enrich : log2 enrichment of the relevant token/literal in the arm vs baseline	any_anchor_token_enrich ≥ 0.8
i2s_vp_WLWL	(inconclusive)	branches with no validated label	33	—	—	—	—
i2s_vp_WLW_	i2s_literal_operand_substitution	I2SRandReplace splices the exact logged comparand literal into the fixed gate offset; VP's CMP_MAP Hamming gradient cannot reach the exact value within budget	3	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 1.0
i2s_vp_WLW_	(inconclusive)	branches with no validated label	4	—	—	—	—
i2s_vp_WL_L	i2s_exact_literal_constant_gate	cmplog plants an exact fixed-offset literal/magic constant the VP gradient cannot climb to	6	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	signed_target_enrich ≥ 1.0 AND decoy_enrich ≤ 0.0
i2s_vp_WL_L	i2s_derived_signed_operand_construction	cmplog constructs a DERIVED negative/range value across parser-routing bytes (no stored literal)	0	operand_enrichment	signed_target_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	no_gate_signature == true (operand_enrichment returns no fixed-offset literal); hypothesis confirmed by the diagnostic ABSENCE of literal enrichment under the i2s-pro shape, failing to inconclusive...

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Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp_WL_L	(inconclusive)	branches with no validated label	1	—	—	—	—
i2s_vp_WWLW	vpc_corpus_inflation_interference [41]	vpc's combined I2S+VP corpus admission inflates and collapses composition, starving the single-technique-found path that both cmplog-alone and value_profile-alone sustain	0	corpus_size_ratio	vpc_corpus_size, cmp_corpus_size, vp_corpus_size, vpc_target_value_fraction	vpc_corpus_size : value_profile.cmplog saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat cmp_corpus_size : cmplog/I2S saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat vp_corpus_size : value_profile.saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat vpc_target_value_fraction : (derived metric — see hypothesis prediction/params)	$\text{vpc_corpus_size} / \max(\text{cmp_corpus_size}, \text{vp_corpus_size}) \geq 1.3$ AND $\text{vpc_target_value_fraction} < 0.5 * \min(\text{cmp_target_value_fraction}, \text{vp_target_value_fraction})$
i2s_vp_WWLW	(inconclusive)	branches with no validated label	2	—	—	—	—
i2s_vp_WWWL	literal_value_gate_cmp_solvable	A fixed-offset literal value gate (FOURCC/tag/enum/single byte) is solved by any CMP-aware fuzzer (I2S substitution or VP Hamming gradient) while naive's blind havoc cannot place the literal	7	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{signed_target_enrich} \geq 1.0$ AND $\text{decoy_enrich} \leq 0.0$
i2s_vp_WWWL	(inconclusive)	branches with no validated label	7	—	—	—	—
i2s_vp_WW_L	parallel_fixed_offset_literal_gate	A fixed-offset 1-4 byte VALUE literal/enum gates the branch; both I2S (direct substitution) and VP (Hamming/prefix gradient) drive the same gate byte(s) to the target value, so both resolving corpora are enriched in the target byte at the gate offset relative to naive.	4	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{cmplog.signed_target_enrich} > 0.5$ AND $\text{value_profile.signed_target_enrich} > 0.5$
i2s_vp_WW_L	parallel_structural_accumulation_gate	The gate is NOT a single fixed-offset literal but a structural/length/cascade condition (repeated integer-token accumulation, or a multi-comparison chunk-validation cascade). Both I2S and VP help by accumulating/clearing many comparisons, so the resolving seeds are larger / carry more target tokens than the blocking seeds — there is no single alignable gate byte.	0	joint_necessity	size_lift, tag_lift	size_lift : winner/loser median seed-size ratio (assembly extent the gradient grows) tag_lift : $\text{winner_tags} - \text{loser_tags}$ (structural-token advantage)	$\text{size_lift} \geq 1.3$ AND $\text{tag_lift} \geq 1.3$
i2s_vp_WW_L	(inconclusive)	branches with no validated label	5	—	—	—	—
i2s_vp_W_WL	i2s_fixed_offset_literal_substitution	I2S enriches the cmplog corpus with the exact CMP-target byte(s) at the gate offset (literal substitution), which naive's blind havoc cannot place	37	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{signed_target_enrich} \geq 1.0$
i2s_vp_W_WL	(inconclusive)	branches with no validated label	19	—	—	—	—
i2s_vp_W__L	i2s_literal_substitution_gate	A fixed-offset literal CMP gate that I2S satisfies by splicing the logged operand byte(s) into the input; cmplog's corpus is enriched for the target literal at the gate offset where naive's is not	1	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{gate_signature} == \text{true}$ AND $\text{signed_target_enrich} \geq 1.0$
i2s_vp_W__L	i2s_relational_collision_gate	A relational / structural gate (hash-bucket chain growth) with NO single byte-traceable operand: I2S substitutes name-equality operands to manufacture COLLIDING entries, so the decisive feature is multiplicity of related names, not one literal at one offset	3	operand_enrichment	signed_target_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	$\text{gate_signature} == \text{false}$
i2s_vp_W__L	(inconclusive)	branches with no validated label	3	—	—	—	—
i2s_vp_LWL	i2s_exact_literal_gate	vpc resolves an exact-literal equality gate because its I2SRandReplace splices the logged comparand literal into a fixed input offset, which value_profile's Hamming/CMP_MAP gradient cannot land	78	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{signed_target_enrich} \geq 1.0$
i2s_vp_LWL	(inconclusive)	branches with no validated label	12	—	—	—	—
i2s_vp_LW__	i2s_multibyte_literal_splice_vp_gradient_blind	vpc's I2S enriches the exact multibyte literal in its corpus; value_profile's gradient does not	0	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted) decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys)	$\text{signed_target_enrich} > 0$ AND $\text{decoy_enrich} < 0.5$
i2s_vp_LW__	(inconclusive)	branches with no validated label	3	—	—	—	—
i2s_vp_WLW	decoy_overcommit_wrong_literal	vpc's I2S substitution pins the corpus to a WRONG decoy literal (sibling case / enumerated case / valid-range boundary), flooding the loser corpus with a value that blocks the target arm	6	operand_enrichment	decoy_enrich, signed_target_enrich	decoy_enrich : same log2 ratio for the loser/decoy gate-byte (>0=I2S corpus pinned to decoys) signed_target_enrich : log2 freq-ratio of the winner's target gate-byte in the I2S-arm corpus vs naive (>0=enriched, <0=depleted)	$\text{decoy_enrich} > 0.8$ AND $\text{signed_target_enrich} < 0.3$
i2s_vp_WLW	(inconclusive)	branches with no validated label	6	—	—	—	—
i2s_vp_WL__	vp_gradient_value_distance	value_profile's CMP_MAP gradient closes the value-distance to the case constants that vpc does not reach	0	value_distance_reached	vp_min_distance, vpc_min_distance, distance_gap	vp_min_distance : min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) vpc_min_distance : min Hamming/value distance from value_profile.cmplog seeds to the gate literal (0 = gate solved) distance_gap : winner — loser gap in value/distance closed toward the gate	$\text{vp_min_distance} < \text{vpc_min_distance}$ AND $\text{distance_gap} \geq 2$
i2s_vp_WL__	(inconclusive)	branches with no validated label	1	—	—	—	—

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Shape	Cluster	Cluster description	#br	Tool(s)	Metric(s)	Metric meaning	Verify rule
i2s_vp__WWL	vp_cmpmap_gradient_distance_closure	VP's CMP_MAP distance-bucket feedback admits seeds that progressively CLOSE the byte-distance to the gate's comparison target (an exact multibyte literal, a grammar-token sequence, or a multi-field format constraint), so the VP-resolving corpus sits strictly closer to that target than the naive-blocking corpus — a continuous gradient naive's edge-only feedback cannot sense.	5	value_distance_reached	vp_min_distance, naive_min_distance, distance_closure_ratio	vp_min_distance: min Hamming/value distance from value_profile seeds to the gate literal (0 = gate solved) naive_min_distance: min Hamming/value distance from naive seeds to the gate literal (0 = gate solved) distance_closure_ratio: fraction of the value gap the winner closes (vs the loser/naive residual)	naive_min_distance > 0 AND distance_closure_ratio ≥ 0.5
i2s_vp__WWL	vp_cmpmap_gradient_depth_reach	For gates with NO single fixed operand (digit-range loops, derived-integer arithmetic, deep multi-token parse reductions, multi-field format integrity), VP's gradient is visible as PARSE/STRUCTURE DEPTH: the VP-resolving corpus reaches a deeper parse state / longer in-range run / more-complete valid structure than the naive-blocking corpus, which stalls shallow.	0	depth_reach	vp_depth, naive_depth, depth_lift	vp_depth: max loop/recursion/path depth reached at the gate by the value_profile arm naive_depth: max loop/recursion/path depth reached at the gate by the naive arm depth_lift: max loop/recursion/path depth reached at the gate by the arm	depth_lift ≥ 1.5 AND naive_depth < gate_depth_threshold
i2s_vp__WWL	(inconclusive)	branches with no validated label	9	—	—	—	—
i2s_vp__W_L	vp_basin_escape	VP CMP_MAP steers AWAY from a dominant magic-value basin: there is no single target literal — the true side wants ANY value but the corpus-dominant magic. VP-resolving seeds DEPART the basin (multi-modal: spread over several non-magic format magics), while naive-blocking seeds are pinned at the basin value.	3	value_distance_reached	naive_basin_lock_fraction, vp_basin_lock_fraction, vp_value_modality, basin_departure_lift	naive_basin_lock_fraction: naive fraction of seeds locked in the wrong value basin (gradient trap) vp_basin_lock_fraction: value_profile fraction of seeds locked in the wrong value basin (gradient trap) vp_value_modality: # distinct value modes value_profile explores at the gate basin_departure_lift: fraction of seeds locked in the wrong value basin (gradient trap)	basin_departure_lift > 0 AND naive_basin_lock_fraction ≥ 0.7 AND vp_value_modality ≥ 3
i2s_vp__W_L	vp_structural_assembly_gradient	VP CMP_MAP closes DERIVED integer comparisons (table offsets/lengths computed from structural fields), not a fixed input literal; VP-resolving seeds carry a larger/more-valid parsed structure built from padding, with no fixed-operand distance signal.	0	struct_size_ratio*	vp_struct_size, naive_struct_size, struct_size_lift, vp_residual_distance, naive_residual_distance	vp_struct_size: value_profile median structural sub-record size at the gate naive_struct_size: naive median structural sub-record size at the gate struct_size_lift: winner/loser structural sub-record size ratio vp_residual_distance: value_profile remaining distance to the gate value after best approach naive_residual_distance: naive remaining distance to the gate value after best approach	struct_size_lift ≥ 1.3 AND distance_close_lift < 0.25 * naive_residual_distance
i2s_vp__W_L	vp_direct_value_gradient	VP CMP_MAP climbs a monotone distance gradient toward a FIXED target value (single literal / switch constant / masked-equality / terminal byte of a chain); VP-resolving seeds reach a smaller residual distance to that target than naive-blocking seeds, which stall at a fixed distance.	0	value_distance_reached	vp_residual_distance, naive_residual_distance, distance_close_lift	vp_residual_distance: value_profile remaining distance to the gate value after best approach naive_residual_distance: naive remaining distance to the gate value after best approach distance_close_lift: winner — naive distance closed toward the gate	distance_close_lift > 0 AND vp_residual_distance < naive_residual_distance (after sense-inversion for mismatch gates) AND abs(distance_close_lift) ≥ 0.25 * naive_residual_distance
i2s_vp__W_L	(inconclusive)	branches with no validated label	8	—	—	—	—
mopt_mutation_LW	mopt_scheduling_keyword_assembly_deficit	MOpt operator scheduling under-assembles the gate keyword that naive's havoc reaches	0	joint_necessity, token_count	mopt_tokens, naive_tokens, token_lift	mopt_tokens: # grammar keyword tokens placed per seed by the mopt arm naive_tokens: # grammar keyword tokens placed per seed by the naive arm token_lift: winner_tags — loser_tags (structural-token advantage)	token_lift ≥ 1.5 AND mopt_tokens < naive_tokens
mopt_mutation_LW	(inconclusive)	branches with no validated label	1	—	—	—	—
ngram_coverage_LW	ngram_shallow_corpus_saturation_starves_deep_path [idf]	ngram-4 feedback floods the corpus with near-duplicate shallow-tokenizer seeds, starving energy from the rare deep xmlParseReference->xmlAddChild path that naive reaches	0	corpus_size_ratio	corpus_size_ratio, depth_reach_inv	corpus_size_ratio: saved-corpus size (or I2S-arm/baseline size ratio) — corpus bloat depth_reach_inv: true if the deep path is NOT reached (inverted depth test)	corpus_size_ratio ≥ 1.5 AND depth_reach_inv == true
ngram_coverage_LW	(inconclusive)	branches with no validated label	1	—	—	—	—
ngram_coverage_WL	ngram_sequential_depth_reach	n-gram path-tuple retention lets naive_ngram4 reach deeper sequential-decode path states that plain edge coverage collapses and discards	2	depth_reach	winner_depth, loser_depth, depth_lift	winner_depth: max loop/recursion/path depth reached at the gate by the winner arm loser_depth: max loop/recursion/path depth reached at the gate by the loser arm depth_lift: max loop/recursion/path depth reached at the gate by the arm	depth_lift ≥ 1.5 AND winner_depth > loser_depth
ngram_coverage_WL	(inconclusive)	branches with no validated label	2	—	—	—	—
TOTAL	66 clusters (32 non-empty / 30 distinct mech)	330 val + 292 inconcl = 622 rows (554 distinct br)	330	6 built + 3 unbuilt	built: corpus_size_ratio, depth_reach, joint_necessity, operand_enrichment, token_count, value_distance_reached, struct_size_ratio, value_enrichment	unbuilt: energy_share,	

RQ3 (preview) — resolve-rate per mechanism class on the EXISTING campaign

Validation only, not the RQ3 result: scored the benchmark's OWN LibAFL variants (so pro/anti/joint patterns are expected-by-construction) over the 4 s4 targets (curl/harfbuzz/openthread/sqlite3). Cell = resolve-rate ($\text{resolved_frac} \geq 0.8$); ≥ 0.8 in **bold**; - = 0 branches of the class reached on these targets. The genuine cross-engine table comes from the RQ3 campaign (AFL++/honggfuzz/libFuzzer/libafl). vpc=value_profile_cmplog.

mechanism class	family	#br	naive	fast	cmplog	vp	vpc	grim	ctx	ngram	minim	mopt
decoy_overcommit_wrong_literal	I2S-pro	6	1.00	1.00	0.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00
i2s_exact_literal_constant_gate	I2S-pro	6	0.00	0.00	1.00	0.00	0.00	0.80	0.00	0.00	0.00	0.00
i2s_exact_literal_gate	I2S-pro	78	0.00	0.00	0.03	0.00	1.00	0.42	0.00	0.00	0.00	0.00
i2s_fixed_offset_literal_substitution	I2S-pro	37	0.00	0.08	1.00	0.00	1.00	1.00	0.00	0.00	0.00	0.00
i2s_fixed_offset_literal_value_gate	I2S-pro	73	0.00	0.02	1.00	0.00	1.00	0.94	0.00	0.00	0.00	0.00
i2s_literal_operand_substitution	I2S-pro	3	0.00	1.00	1.00	0.00	1.00	1.00	0.50	0.00	0.50	0.00
i2s_literal_substitution_gate	I2S-pro	1	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
i2s_anti_structural_byte_corruption	I2S-anti	7	1.00	0.86	0.00	1.00	1.00	0.14	0.86	0.86	0.86	0.86
i2s_case_constant_decoy_substitution	I2S-anti	1	1.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00
i2s_decoy_substitution_overfit	I2S-anti	6	1.00	0.83	0.00	1.00	0.00	0.00	0.83	0.83	0.83	0.83
i2s_decoy_substitution_target_depletion	I2S-anti	23	1.00	0.91	0.00	1.00	0.00	0.04	0.91	0.91	0.95	1.00
i2s_energy_diversion_decoy_pin	I2S-anti	2	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00
joint_assembly_depth	joint	7	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
joint_value_precision	joint	15	0.00	0.00	0.00	0.00	1.00	0.08	0.00	0.00	0.00	0.00
vp_cmpmap_gradient_distance_closure	VP-pro	5	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
vp_gradient_climbs_numeric_threshold	VP-pro	4	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
vp_gradient_distance_climb	VP-pro	1	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00
vp_gradient_value_distance_climb	VP-pro	5	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
vp_gradient_value_distance_closed	VP-pro	1	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
vp_pro_cmpmap_gradient_rescue	VP-pro	1	1.00	0.00	0.00	1.00	1.00	0.00	1.00	0.00	0.00	0.00
structural_depth_only	grimoire	2	1.00	1.00	0.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00
structural_token_assembly	grimoire	9	0.50	0.50	0.00	0.50	0.00	1.00	0.50	0.50	0.50	0.50
iteration_depth_threshold	ctx-pro	13	0.00	0.14	0.14	0.43	0.29	0.14	1.00	0.00	0.14	0.00
ctx_depth_inflation_at_gate	ctx-anti	2	1.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	1.00	1.00

The three I2S-anti (“I2S hurts”) mechanisms — description & real-corpus byte evidence

All I2S-anti branches share one measured signature on the campaign corpus: the cmplog (I2S) arm is **depleted** of the byte the winning arm reaches ($\text{signed_target_enrich} < -0.3$, where $\text{signed_target_enrich} = \log_2(\text{target_frac}_{\text{cmplog}}/\text{target_frac}_{\text{naive}})$). They differ in *why* — and the deterministic arbiter separates them by the decisive **shape** (which fuzzers win/lose), the shared $\text{ste} < -0.3$ being the common depletion gate (verify-rule column). Source: `docs/i2s_anti_mechanisms.tex`.

mechanism (shape)	example branch & gate	real-corpus metrics	verify rule (arbiter)	description / reading
1. target_depletion i2s_decoy_substitution_target_depletion (LW_W, n=23)	libpng_3977 — cHRM chunk handler (pngutil.c), single gate byte at input offset 236 . target 0x00 vs decoy 0x65 ('e' from ASCII floats <code>1.0e0"/65.535"</code>).	target_frac 0.089 cmp / 0.157 naive ⇒ $\text{ste} = -0.82$ decoy_frac 0.63 cmp / 0.39 naive ⇒ decoy_enrich = +0.68	shape LW_W and $\text{ste} < -0.3$ and gate present with an identifiable rival operand at the gate offset ($\text{decoy_enrich} > 0$).	A rival real operand wins the harvest. An abundant competing byte (the textual 0x65) sits at the same offset; cmplog logs it from other parser comparisons and I2SRandReplaces it in, <i>pinning</i> 63% of its corpus to the decoy and starving the resolving 0x00 down to 8.9% (naive, never harvesting the decoy, drifts onto the target freely). Two operands competing → I2S captures the wrong one.
2. byte_corruption i2s_anti_structural_byte_corruption (LWWW, n=8)	harfbuzz_5831 — cmplog is the <i>sole</i> loser (naive, vp, vpc all win). No single rival; wide gate (gate_width=29, offsets 71–187).	target_frac 0.21 cmp / 0.38 naive ⇒ $\text{ste} = -0.88$ decoy diffuse: decoy_enrich+2.6 but low-abs; cmplog spread over 211 bytes, largest non-target only ≈ 0.12	shape LWWW (cmplog the <i>sole</i> loser) and $\text{ste} < -0.3$. No identifiable rival decoy needed — depletion alone is the signal.	Blind RandReplace clobbers an assembled byte. Naive havoc <i>assembles</i> the structural byte; cmplog overwrites that exact position with a logged value — no dominant substitute, just scatter. Seed length corroborates truncation: cmplog blocking seeds 10–61 B vs naive resolving 60–188 B (size ratio ≈ 1.7) — cmplog never completes the record. Corruption, not competition.
3. overfit i2s_decoy_substitution_overfit (LWLW, n=7)	harfbuzz_5323 — hb_script_get_horizontal_direction:594, case HB_SCRIPT_THAANA. The gate is not an input byte : it resolves iff the decoded text carries Thaana codepoints (U+0780–07BF) — a <i>semantic</i> property. Decoy = first OpenType table tag at [12:16] (GPOS/GSUB/MATH).	target_frac 0.082 cmp / 0.508 naive ⇒ $\text{ste} = -2.64$ (strongest in the family) decoy_frac 0.51 cmp / 0.0004 naive ⇒ decoy_enrich = +10.4	shape LWLW (cmplog & vpc lose) and $\text{ste} < -0.3$, with the decoy a structural “stay-valid” literal ($\text{decoy_enrich} \gg 0$).	A validity trap that starves a semantic gate. cmplog <i>reaches</i> the branch every trial but can't <i>produce</i> exotic-script text: its I2SRandReplace keeps re-asserting structural literals (font table tags), spending 51% of its corpus on valid ASCII/Latin fonts and starving the Thaana script the gate needs. <i>Caveat</i> : the byte stat is a structural <i>correlate</i> of the semantic gate, not the gate itself — the deterministic tool's honest ceiling (see M3 note in the source).